

PORTFOLIO CASE STUDY

Petalcrumb

Cake Studio

Bespoke celebration cakes, hand-finished in London — built end-to-end as a full-stack portfolio case study.

Live · petalcrumb-cake-studio.vercel.app

Code · github.com/andrehuang13-eng/petalcrumb-cake-studio

ANDRE HUANG · 2026 · CONCEPT PROJECT



End-to-end full-stack — built and deployed.

What it is

A complete website for a fictional boutique cake studio: a database-driven customer catalogue, a structured enquiry flow with email notifications, and a login-gated admin CMS the owner can use to manage every piece of content.

My role

Solo full-stack developer — product framing, UI design system, backend, database schema, auth, deployment, observability.

Why concept

No real client. I drew up an imaginary brief with realistic constraints and treated the build as if a real bakery owner had hired me, so every decision had to justify itself.

Stack

Framework · Next.js 16 (App Router, RSC, Server Actions)

Language · TypeScript, React 19

Data · Prisma + PostgreSQL (Neon)

Styling · Tailwind v4 · light + dark mode

Auth · jose JWT + bcryptjs (custom session)

Validation · Zod (shared client + server)

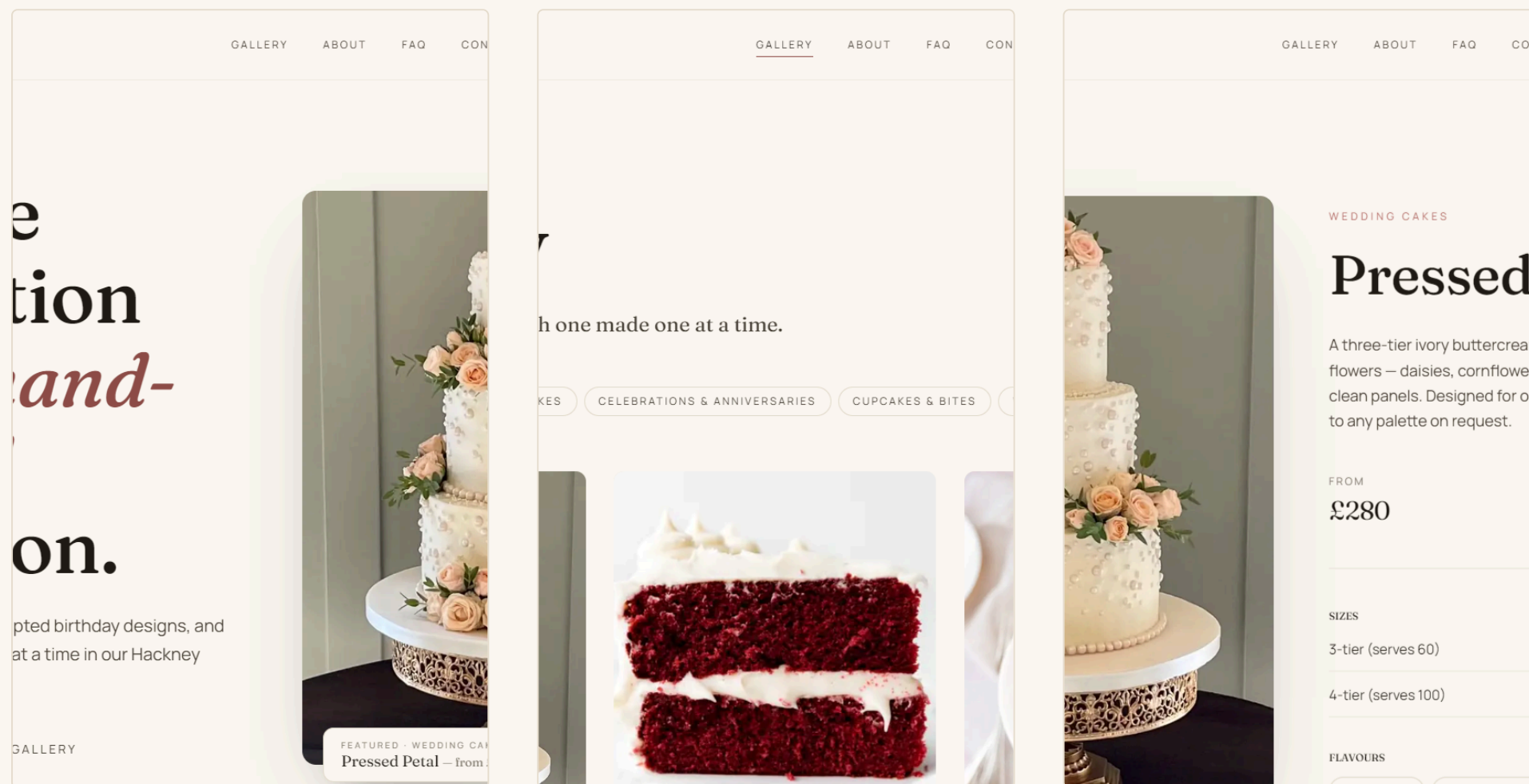
Email · Resend

Storage · Vercel Blob (image uploads)

Host · Vercel (CI auto-deploy on push)

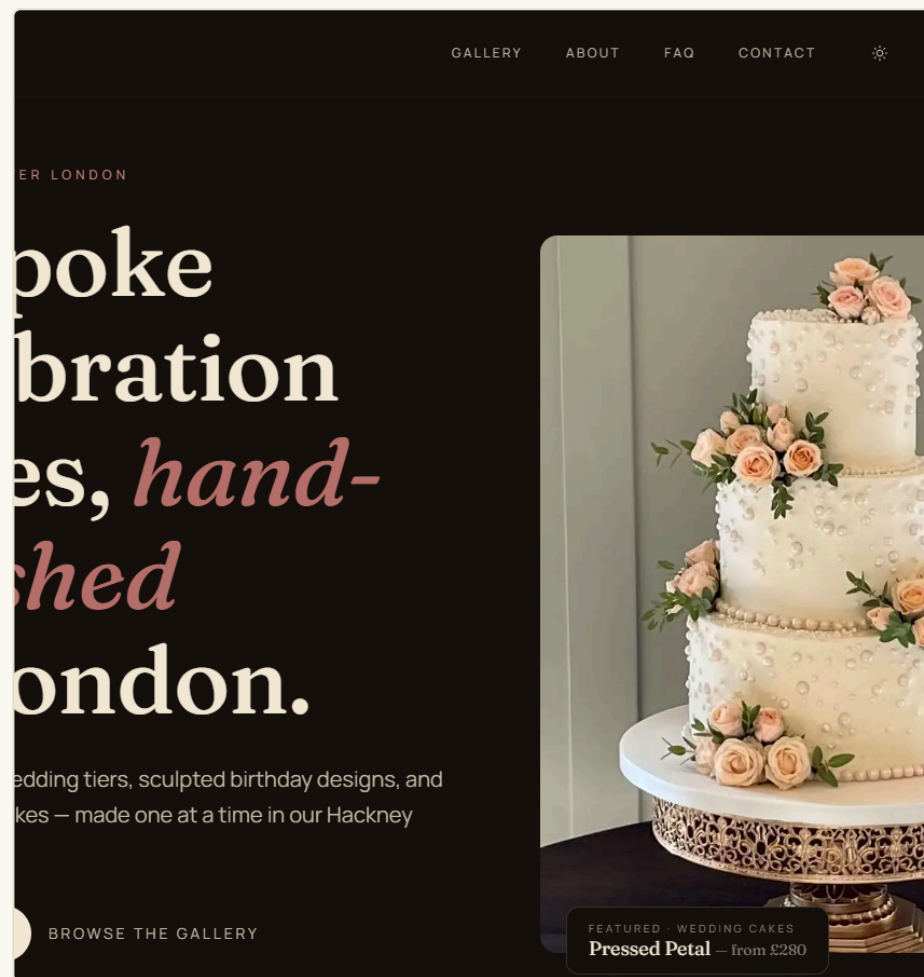
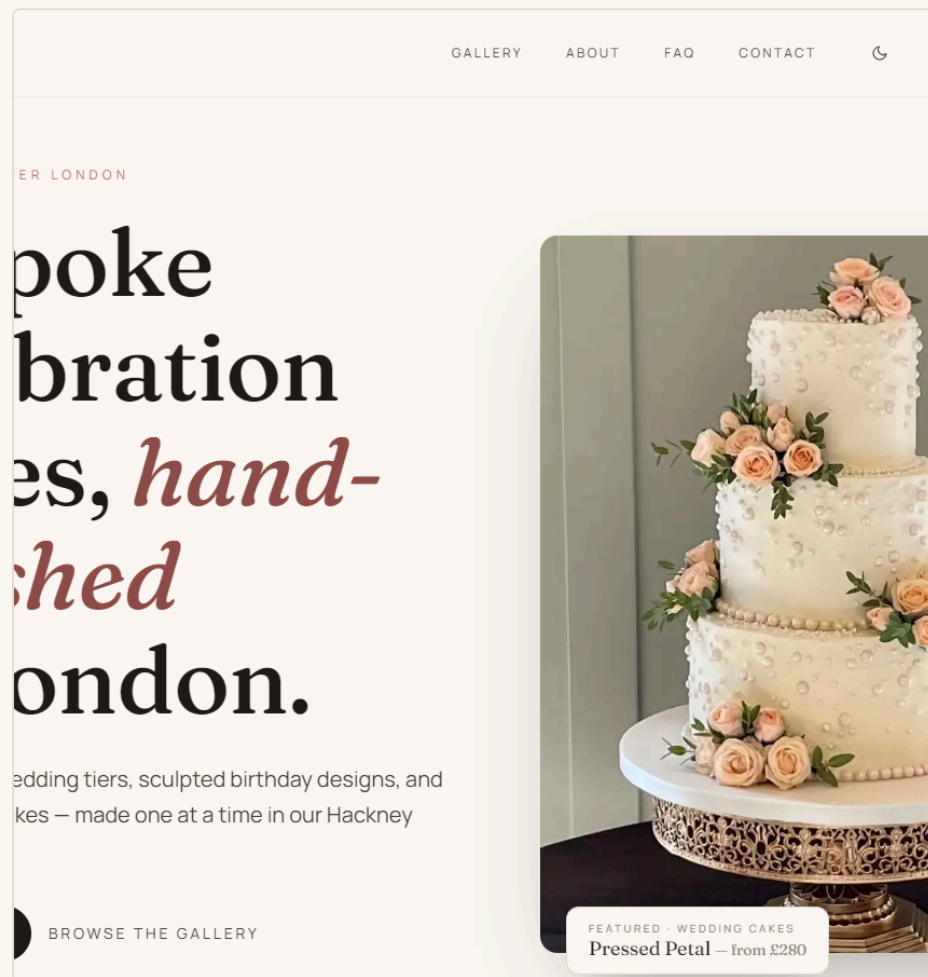
Editorial design, served from the database.

Catalogue + cake detail + enquiry flow all render from PostgreSQL. The "Request this design" CTA pre-fills the form with the visited cake.



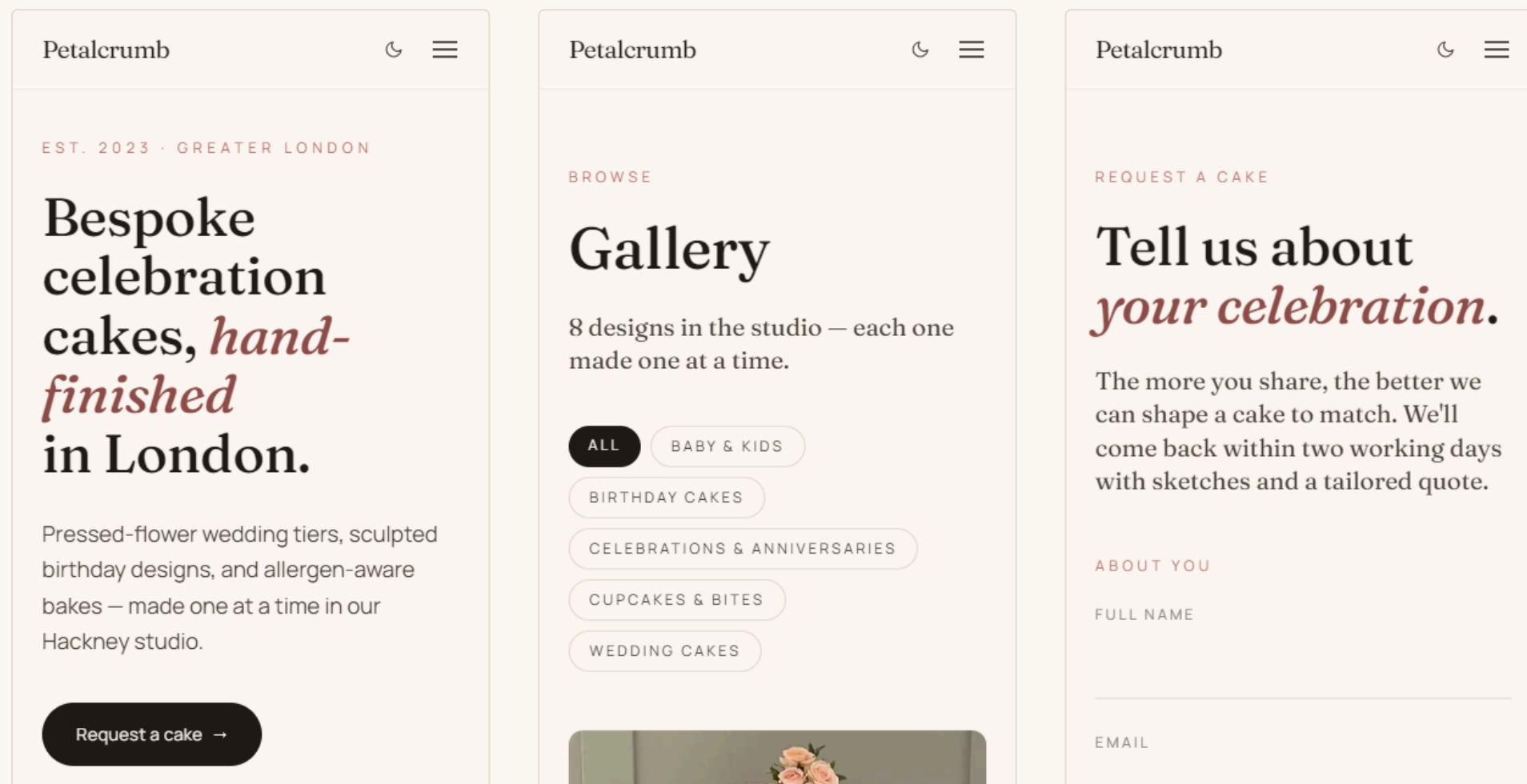
Same design system, *two palettes*.

Implemented via CSS variable swap on a single `.dark` class — zero markup changes between modes. Persisted via cookie + system-preference fallback, FOUC-free.



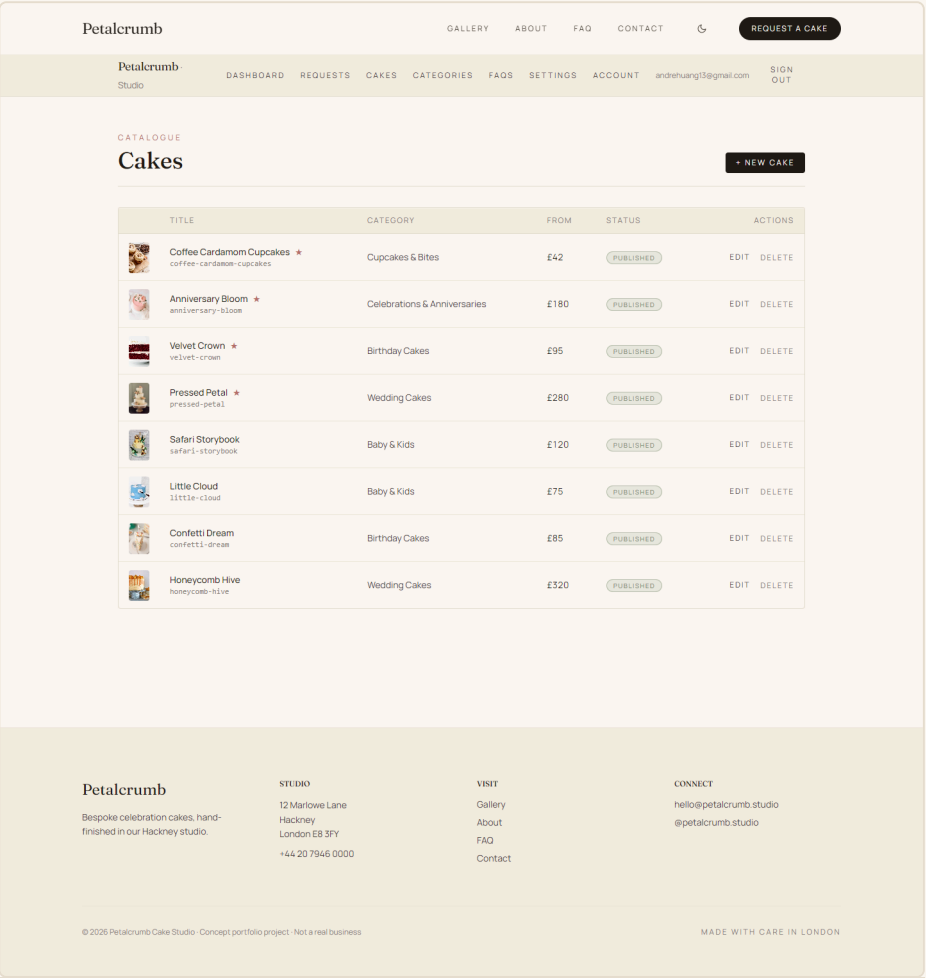
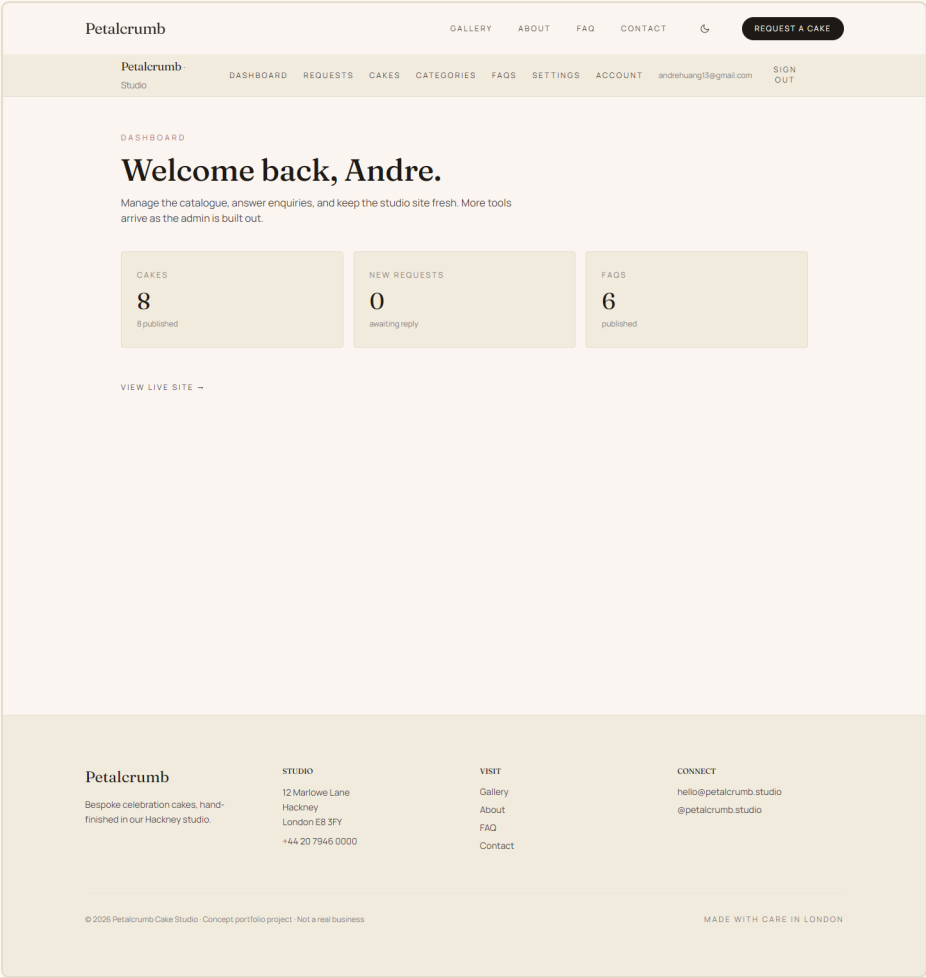
Same care on small screens.

Single column, hamburger menu, full-width cards. Mobile gets every real photo, the marquee, and the form, exactly as desktop does.



The owner runs everything — without touching code.

Login-gated dashboard with full CRUD for cakes (with image uploads), categories, FAQs, settings, and an order-request inbox with status triage.



Real workflow: core fields, sizes, flavours, images.

One form combines all data points. Images upload to Vercel Blob (or paste an external URL). Sizes are an inline sub-form. Delete is guarded — categories with referenced cakes won't delete.

Petalcrumb

GALLERYABOUTFAQCONTACT

REQUEST A CAKE

Petalcrumb Studio

DASHBOARDREQUESTSCAKESCATEGORIESFAQSSETTINGSACCOUNTandrehuang13@gmail.comSIGN OUT

← CAKES

CATALOGUE

Edit · Pressed Petal

VIEW LIVE →

TITLE

Pressed Petal

SLUG

pressed-petal

Leave blank to generate from the title.

CATEGORY

Wedding Cakes

DESCRIPTION

A three-tier ivory buttercream cake finished with edible pressed flowers — daisies, cornflowers, and rose petals set against ribbon-clean panels. Designed for outdoor summer weddings, but adapted to any palette on request.

Decisions worth calling out.

Auth without a library

jose-signed JWT in an httpOnly cookie, bcryptjs hashes. Built per Next 16's own auth guide — chose this over Auth.js v5 because v5 was still beta and Next 16 renamed middleware to proxy.

One Zod schema for both sides

The order form validates with the same schema on client (React Hook Form) and server (Server Action). Drift is impossible by construction.

Email survives serverless

Resend send is awaited inside the action so it actually delivers — the original fire-and-forget was killed by function freeze. Failures are swallowed and logged so the user never sees them.

Vercel build is reproducible

postinstall: prisma generate fixes the stale-client cache trap after schema changes. Blob host is allowlisted in next.config. Dynamic OG + Twitter image generated by next/og.